A

Project Report

on

"AI Powered Customer Sentiment Analysis for E-Commerce"

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Mahatma Gandhi Mission's College of Engineering, Nanded (M.S.)

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A Project Report on

"AI Powered Customer Sentiment Analysis for E-Commerce"

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Certificate

This is to certify that the project entitled

"AI Powered Customer Sentiment Analysis for E - Commerce"

being submitted by ***Ms. Amruta Patil, Ms. Suhani Londhe, Ms. Saloni Navgire and Ms. Shejal Waghmare*** *to the Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by them under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

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ABSTRACT

The project “AI-Powered Customer Sentiment Analysis for E-Commerce Website” is designed to analyze customer reviews and classify them into positive, neutral, and negative sentiments using Artificial Intelligence. The system allows users to enter a product URL, from which real-time reviews are collected using web scraping techniques. These reviews are then preprocessed and converted into numerical features using TF-IDF for analysis.

The system includes an interactive frontend developed using HTML, CSS, and JavaScript, providing user-friendly pages such as login, registration, input forms, and dashboards with animations and visual insights. Multiple Machine Learning models such as Naive Bayes, SVM, Logistic Regression, Decision Tree, and KNN, along with Deep Learning models like CNN, LSTM, BiLSTM, and BERT, are trained and evaluated. A Hybrid model (CNN + SVM) is also used to improve performance, and the best model is saved for prediction.

The backend is implemented using Flask, where results are displayed on a dashboard with insights, trends, and sentiment distribution. Additional features include emotion detection, smart reply generation, and downloadable reports, with a MongoDB Atlas database used to store user data and analysis history. This system helps businesses understand customer feedback and make better decisions.

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INTRODUCTION

In the current world, e-commerce transactions have been made integral parts of our lives. Online transactions are preferred by people since they are convenient, fast, and offer a wide variety of choices. The only problem is the stiff competition witnessed by organizations in the field of e-commerce. This stiff competition makes it imperative for organizations to understand the opinions and views of the customers. Customer views are expressed through product reviews, ratings, comments, and social media interactions. Product reviews consist of useful customer opinions about the product or service. However, one major problem with product reviews is that they occur in massive numbers and unstructured formats. Artificial Intelligence (AI) and Machine Learning (ML) can help solve this problem.[1]

The central objective of this project is to create an AI-based system that will examine customer reviews obtained from e-commerce sites and categorize those into various sentiments. Such a system will be able to determine how happy or unhappy the customers are with a product. Based on this information, a company will be able to improve its products, services, and customer experience. Customer feedback plays an important role in e-commerce organizations for making decisions about products and improving services. Sentiment analysis enables such firms to make effective use of this feedback and analyze customer behavior and sentiments. The project aims to build an automated machine learning model for performing sentiment analysis on customer reviews. Text preprocessing and feature extraction methods (TF-IDF) and classifiers (Logistic Regression/Other ML Algorithms) will be employed for developing the solution. The dataset will be made up of customer reviews gathered from e-commerce websites.

The sentiment analysis process consists of several important steps. At first, the information is gathered from various resources, which might include product reviews and websites. Next, the data is preprocessed and cleaned by removing unnecessary parts, such as stop words, punctuation, and any special symbols. Afterwards, the text is transformed into numbers that could be read by algorithms. Finally, the algorithm is trained for sentiment classification of each review. The described process enables one to derive valuable information from the raw data. One of the main benefits of the project is its automatic nature. Instead of spending hours on reading countless reviews, the system performs this task instantly, saving lots of effort and time. Moreover, it can be easily combined with dashboards and visual representations in the form of graphs.

Sentiment analysis is essential in terms of making decisions in real time. In addition to responding to consumer problems and improving weaknesses, such technology will help increase the satisfaction of customers. When there are negative reviews regarding the features of some product, the company can react promptly to this problem. In recent times, artificial intelligence-based sentiment analysis has been recognized as an effective means of business success. Thanks to AI, businesses will be able to understand customers' feelings better and create appropriate marketing and products development strategies. Thus, apart from the focus on sentiment analysis, the project proposes a comprehensive system covering such aspects as processing data, training models, prediction, and results visualization. Such a project shows how AI may be effectively applied in practice, especially in the sphere of e-commerce. To sum up, it may be concluded that the proposed system for analyzing customer sentiments with the use of artificial intelligence is very efficient and useful.

1.1 Background

In the last decade, there has been a tremendous change in how customers purchase goods. Old shopping modes, involving the presence of the customers at physical shops, have been taken over by modern online shopping sites. Due to the advancements made by technology, the e-commerce sector is currently one of the rapidly expanding sectors in the world. Sites such as Amazon, Flipkart, among others, offer customers the chance to browse and buy products.

With an increasing number of online users, there will definitely be more data being produced by customers as well. Customers leave reviews and ratings after buying any kind of product. Reviews could range from quality of the product, service provided by delivery, price, usability, and customer satisfaction. This information will become very useful to the organization since they give us the direct voice of the customer. However, there will be a big challenge ahead. Millions and billions of reviews are created by customers daily. If an item or product becomes popular, they might generate millions of reviews within a very short time frame. As a result, the number of reviews that need to be read by organizations will be enormous. Another point that should be considered here is that customer feedback will not always be simple and clear. In addition, people might use a variety of languages or terms when providing their feedback to companies.

Previously, companies were dependent on basic tools such as using ratings (such as one to five stars) for gauging customer satisfaction. Though ratings offer a sense of satisfaction, there is no information available regarding the opinion of the client in terms of the text of the review itself. There might be times when a consumer gives a good rating but states a serious problem in the review text itself. All this is missed by conventional techniques. To address such limitations, the tool of sentiment analysis was developed. Sentiment analysis is an approach where the mood of the text is analyzed to find out whether the message is negative, positive, or neutral. This method is part of the Natural Language Processing approach which is a field of artificial intelligence that deals with natural languages.

Initially, in the early stage, the sentiment analysis task was solved using rule-based approaches. In these approaches, pre-defined sets of positive and negative words are used for labeling reviews as positive or negative. Examples of positive words include "good," "excellent," "amazing" and examples of negative words include "bad," "poor," and "worst." But such rule-based approaches suffered from a number of disadvantages. Such techniques failed to capture the context of sentences, any sarcasm present in the review, or complex structure of sentences. Consider, for example, the sentence "the product is not bad" will be classified as negative, although it does not convey a negative meaning. With advancement in technology, now the sentiment analysis task has been successfully carried out using machine learning methods. These machine learning models make use of previous experiences to learn new facts. To train these models, huge datasets consisting of positively or negatively labeled reviews can be used.

Another advance was achieved through the implementation of advanced techniques such as feature extraction techniques, for example, using TF-IDF (term frequency-inverse document frequency), which allows to define relevant terms within the document. The features obtained through such techniques are used as an input in the process of training different machine learning algorithms, including Logistic

Regression, Naïve Bayes, and Support Vector Machines. These techniques greatly helped to improve the efficiency of sentiment analysis tools. Recent times saw advances in deep learning and neural networks applied to the process of sentiment analysis. For instance, RNNs (Recurrent Neural Networks) and CNNs (Convolutional Neural Networks) are able to capture complex patterns in text and understand context better.

In the realm of e-commerce, the significance of sentiment analysis has gained much importance for businesses. Sentiment analysis enables companies to know what customers feel, detect any issues among consumers, and refine their services accordingly. For instance, if a majority of consumers feel that there are delays in deliveries, then the company will work towards fixing the problem at once. Likewise, positive sentiments will help companies recognize their strengths and capitalize on them in the right manner. The other crucial factor is making decisions. Using sentiment analysis, businesses will be able to make better decisions using facts rather than assumptions. This will not only help them enhance consumer satisfaction but also increase sales and build brand loyalty.

Even with all its benefits, sentiment analysis is faced with some limitations. Sarcasm, mixed emotions, and multilingual texts pose difficulties. Take for instance a statement such as "It is a great product, but the delivery service is very bad." Both sentiments are expressed within the sentence. Such situations call for an upgrade in approaches for analyzing sentiments from text. In our project, we seek to solve the problems that come with sentiment analysis by developing a tool that will use machine learning algorithms to do the job. The project will ensure efficient processing and classification of textual data.

Introduction to this project explains the increasing need to analyze the customer feedback in the present era of digitalization. As e-commerce grows and expands, there would be an increased requirement of an intelligent system to assess customer feedback. The project proves that Artificial Intelligence is capable of solving practical issues.

1.2 Problem Statement

In today's rapidly advancing digital world, there is an ever-increasing necessity for people to buy various items and services through ecommerce platforms. In the process, buyers depend highly on reviews before buying anything. Reviews play a crucial role in providing customers with information regarding the quality and reliability of the item. On the other hand, it is equally useful for businesses because it enables them to get a sense of how well the products perform from the point of view of the customers. The most significant problem that has resulted in this regard is the massive amount of review information that is generated on a daily basis. Millions of reviews are posted every day by popular ecommerce websites.

In addition to this, another important problem is that the reviews of customers are written in an unstructured text format. Contrary to the numeric data, the processing of text data is a challenging task as well. The review of the customer might contain slang, emoji, informal words, errors in spellings, and even multiple languages. Even a word can be used in a completely different sense based on the context in which it appears in the review. In this way, the statement "This product is not bad" implies that the reviewer likes the product despite using the word "bad." Therefore, the traditional practices being adopted by the organizations, i.e., analyzing stars from 1 to 5, are not enough. The client might provide a positive rating but at the same time point out some problems with his/her experience during writing a review. Likewise, a negative rating does not necessarily mean that all the experience of the client was bad. Consequently, an analysis based solely on ratings results in making wrong decisions and drawing false conclusions. Reviews' analysis by hand takes too much time and is inefficient because it involves a lot of errors and inaccuracies. At the same time, it is impractical for companies to employ specialists who will analyze the whole mass of data.

One such challenge is that of real-time analysis. In the field of e-commerce, the preferences of the consumers are always changing at a rapid pace. It is therefore imperative that companies act on customer feedback instantly in order to remain ahead in the competition. Without an automated system, however, real-time analysis of consumer sentiment proves to be difficult.

Moreover, gauging the sentiment of the customers is sometimes difficult to determine. Customer reviews can have mixed feelings where a customer can say both positives and negatives in a single review. For instance, "Product quality is great but the delivery was too late." In such a case, it is hard to gauge the actual sentiment of the customer. Simple approaches that are already being used find it difficult to cope with this situation. Moreover, there is the problem of determining whether the review itself is a false one. Reviews sometimes can be misleading and can contain biases against a certain product or service.

As a result of these issues, it becomes challenging for businesses to exploit fully the potential value lying within consumer reviews. This will, in turn, negatively impact their capability to refine their products, improve customers' satisfaction, and help them make better business decisions. Without adequate analysis of these consumer reviews, businesses might miss out on critical problems as well as areas where they excel. Therefore, there is a need for an automated and intelligent system capable of analyzing consumer reviews fast and efficiently. The ideal solution would be a system that can interpret and categorize sentiment as positive, negative, or neutral.

The suggested approach involves the creation of an AI-based customer sentiment analysis tool utilizing both machine learning algorithms and NLP methods. This tool will enable automated analysis of customer feedback, extracting useful information about the attitudes of customers to the products/services provided by the company.

Additionally, the system needs to consider that text data could take various forms, such as non-formal text and diverse writing styles. Through methods like text preprocessing, feature extraction, and the use of classifications algorithms, the system will be able to learn the patterns in the available data and make informed decisions accordingly. Ultimately, what the project seeks to achieve is helping firms have better insight into their customers. By knowing their opinions, either positive or negative, businesses will know where they need to improve their goods and services.

1.3 Significance of the Project

In the current age of information technology, data has turned out to be the most precious resource for every company. In particular, in the case of the e-commerce sector, customer reviews, comments, and rating are the most important elements to help companies know what customers want. But in this scenario, because of the enormous size and unorganized nature of this type of data, it becomes quite tough to process this data manually. This is where the importance of the current project emerges. The main purpose of this project is that it helps analyze customer reviews automatically. It enables the businesses to save their time and energy by not having to read through tons of reviews manually.

One of the important significances of the research is that it will aid companies in understanding their customers better. Reviews from customers provide useful data regarding the quality of the product, the process of delivering it, the price of the product, and the overall customer satisfaction with the product. By applying machine learning algorithms on the reviews, businesses can understand the problems and strengths of their products. For instance, when customers report a problem repeatedly, the business can resolve it immediately.

Furthermore, this project will also help in enhancing decision-making for any business. Traditionally, business decisions have been made by making assumptions or on the basis of less information. In contrast, the use of sentiment analysis will allow the business organizations to make data-based decisions. It helps in making better decisions, resulting in better products and increased customer satisfaction. Moreover, an important benefit offered by the use of sentiment analysis in business organizations is that of real-time analysis. In a competitive environment, real-time analysis is quite important for the business organizations. It helps in detecting the customer sentiments instantly, and action can be taken immediately.

The project is equally important when considering improving customer satisfaction. Once a business understands its customers' requirements, it would be in a better position to deliver quality services to them. Satisfaction and loyalty are key factors to any successful company and hence the need for such a project. In terms of technology, the project offers an example of how AI and ML could be put into practice in dealing with real-life situations. For instance, NLP, text processing, and classifications are some of the ways in which valuable data could be extracted from the structured text.

Another key importance is scalability. This means that the system will be able to process big volumes of data. Therefore, this software product can be used both by small companies and by big online retailers. Moreover, even if the number of users and reviews rises, the efficiency of the system will not decrease much. This research project has helped the firm grow and compete on the market. Due to analysis of the customers' feelings, businesses will be able to develop their products better and receive positive feedback.

Moreover, this project can be further expanded and incorporated into other platforms such as dashboards and analytical platforms. In doing so, organizations will have an opportunity to present their findings in the form of charts and graphs which makes it easier for them to interpret their findings. Such platforms can also be used for other purposes such as social media monitoring and brand analysis.

1.4 Objectives of the Project

In the context of the current project, the overall aim is to implement an intelligent system which can analyze reviews written by customers for products or services sold on e-commerce sites and determine the sentiment of these customers. In the modern world, it is quite common to see customers providing feedbacks about their shopping experience on different platforms; however, since the volume of data generated by these people is high, manual analysis does not seem to be the ideal solution. Thus, a main objective of this project is to design a highly efficient and intelligent system capable of analyzing reviews and generating useful information out of these. A key objective is to construct a website that automatically determines the sentiment of customer reviews.

Another critical task will be the processing and cleaning of unstructured text data. In many cases, customer reviews are not organized and may consist of redundant elements such as punctuation marks, special symbols, and stop words. It is one of the main tasks of this project to preprocess this data to facilitate the analysis. Clean data is vital in ensuring the correctness of the results provided by the machine learning algorithm. In addition, text needs to be converted into numbers to enable machine learning algorithms to analyze it. Tools such as TF-IDF can be used for this purpose.

An important goal of the project is to train and deploy a machine learning algorithm capable of predicting the sentiments expressed by customers through their reviews. Machine learning algorithms like logistic regression or others can be applied to extract patterns from the data. Training involves building the model using a training set of reviews and testing its performance. In addition, an important goal of the current project is to increase the accuracy of sentiment prediction. The current project concentrates on choosing an appropriate model and optimizing it to attain high levels of accuracy. Metrics including accuracy, precision, recall, and F1-score will be considered during the evaluation stage to make sure that the final results obtained with the help of the developed solution will be reliable. Besides, it should be noted that the project will provide important insights into decision-making. Analyzing sentiments expressed by customers allows us to make better decisions regarding our product or service offerings.

Another purpose is to create a user-friendly interface that enables users to provide their feedback and predictions of sentiments. Such a system will be useful for its practical use. Moreover, it can be incorporated in dashboards for easy visualization of the outputs. Yet another purpose of this project includes the management of big data volumes. Because e-commerce sites receive lots of reviews on a daily basis, it is necessary to ensure that the system processes data in an efficient manner without impacting its performance.

An additional goal is the illustration of the practical implementation of Artificial Intelligence and Machine Learning. The purpose of the project is to gain insight into the use of AI techniques to resolve real-world issues. Through this project, one will learn about implementation of NLP, data preprocessing, features extraction, and model training. One can extend the scope of the project for scalability and flexibility in the future. For example, one can improve the current system by building a more sophisticated model, using multiple languages, or connecting it to different platforms such as social media.

1.5 Future Scope

However, the domain of Artificial Intelligence and Machine Learning is always developing, and as a result, there are numerous ways of how such technologies can be improved in terms of building better sentiment analysis systems. In particular, the aim of the current project was to develop an application which would classify customers' reviews according to whether they have a positive, negative or neutral sentiment.

One of the important ways to improve the model in the future includes using advanced deep learning models. At present, logistic regression models and other classification algorithms are applied to perform sentiment analysis. Though these machine learning models offer high accuracy, they lack understanding when it comes to analyzing complex patterns in language structures. Advanced machine learning models such as neural networks, recurrent neural network (RNN), long short-term memory (LSTM), and transformer models can be employed for future applications to predict sentiment better.

Another crucial future scope is the capability of dealing with data that comes in multiple languages. In practical applications, users might provide their reviews using more than one language or use different languages within a single statement. The current system may only be capable of processing text in a single language, probably English. With advancements in the future, the system can become capable of processing text written in multiple languages, including Hindi, Marathi, and other languages. It would make the system more generalized to be used by people from a variety of backgrounds, specifically in India. Another possible application is the development of an aspect-based sentiment analysis model for better insights into user reviews. Currently, the system only focuses on the sentiment expressed in the overall review of the product or service being reviewed. However, each statement in a review may involve a number of aspects. Using the same example, a person may appreciate the quality of a particular product but complain about its delivery services.

Improvement can also be made in the detection of sarcasm and complex emotions. Humans are very clever at playing with language, and they may sometimes be sarcastic or speak indirectly to get the point across. A sentence like “Fantastic job of delivering the product late!” means something entirely different despite saying “fantastic,” which is positive. Future systems can employ more sophisticated methods of Natural Language Processing to handle such complex sentences.

Another improvement for the system includes real-time sentiment analysis. While in the current setup the sentiment analysis takes place on a data set which is already available, in the coming days, the system can be integrated with live data sources. For instance, the business could integrate the system with their e-commerce platform, or any other medium where people leave their feedbacks. Such integration would make the process of analyzing sentiments easier and efficient. One more improvement for the system could include designing an interface through which users will not just get the text result but also see graphical representation of results through graphs, charts, etc. This will make it easier for companies to detect trends, patterns, and other critical insights easily. Companies can know how customer sentiment evolves through time and which products get a lot of negative reviews. The project can also be scaled up to use predictive analytics. Using previous records, the system will be able to predict future trends in terms of customer sentiment. For instance, it may be able to determine whether a particular product will receive negative or positive reviews in the future.

One more useful feature that could be added is the combination of this sentiment analysis with a recommendation engine. The sentiment analysis tool can be used together with a product recommendation system that would give recommendations based on the user's preferences and review information. For instance, positively reviewed items could be recommended to users.

It will also be easier to scale the system and move it to cloud computing. With more data, scalability becomes an essential feature to take into account. The system can be implemented through cloud services, which will allow it to process large amounts of data much faster and reliably. As such, it will become appropriate for use by big e-commerce firms. One way of extending the capabilities of the system would be to include data from multiple sources. While the project might now only consider customer reviews, in the future, it might include additional data such as that from social networks, blogs, forums, and even customer chat sessions.

The system may also adopt feedbacks from customers and keep on learning. Regular updates of machine learning models with fresh data will improve the performance of the system and allow it to adapt to changes in customer behavior and speech patterns. As for future improvements from the standpoint of security and ethics, the project may implement data protection and minimize possible biases. Customer data needs to be protected from unauthorized access, while the algorithm should be free from biases and unfairness. It is critical to implement measures to make the system transparent, safe, and reliable. The project may be turned into a commercial product. For instance, it may be created as a web-based or software application that could be adopted by various companies without any trouble.

Chapter 2

LITERATURE SURVEY

A Literature Survey is a very crucial step in engineering projects or any project at all. This survey helps in understanding technologies and systems that are already available related to the field or subject of interest. For the current AI-Powered Customer Sentiment Analysis project in the domain of e-commerce, the literature survey involves various ways of performing sentiment analysis of customers through NLP, ML, and deep learning techniques.

The reviews left by customers play a pivotal role in the field of e-commerce, nowadays. Many researchers have been trying to analyze customer reviews through automatic techniques. The current study is an extensive review of existing literature in terms of research work done in sentiment analysis, text mining, reviews classification, and decision support systems based on artificial intelligence.

Customer reviews' analysis is extremely critical for businesses since it contributes greatly to improvements in products and services. The process becomes hard when done manually because of the voluminous nature of the data. Numerous automated systems using AI approaches have therefore been established. This survey provides great insight into the automated systems and the development of this project.

2.1 Pang, B. et al. — "Thumbs up? Sentiment Classification using Machine Learning Techniques"

Convenient Name: Pang et al. (2002) – Machine Learning for Sentiment Analysis
Source: EMNLP Conference

The current research paper stands out to be one of the most significant contributions to the body of work in sentiment analysis. In previous approaches to sentiment analysis, simple rule-based mechanisms that relied on lists of words that represented positive or negative sentiments were used. The problem with this approach was that it did not comprehend the underlying structure of the sentence being analyzed.

Pang et al. suggested applying machine learning algorithms for automatic classification of text sentiment into positive and negative sentiments. The authors conducted an experiment with a range of widely used machine learning algorithms like Naive Bayes, Maximum Entropy, and SVMs. In the experiment, the mentioned models were trained with the help of a set of movie reviews which were labeled according to the sentiment expressed in it. Thus, the models learned how to recognize connections and patterns between words and sentiments. One of the major advantages of the study is that it utilizes advanced feature extraction approaches including Bag of Words (BoW). With BoW, each text turns into a number of words without any consideration about grammar or word order. Thus, every word becomes a feature, while the frequency of words helps to represent a document.

Experimentation was done by the authors of the study to compare the efficiency of machine learning systems with rule-based approaches. It was found out that machine learning algorithms performed better as far as accuracy was concerned. This demonstrated that automated sentiment analysis through machine learning is both feasible and superior to its predecessors.

An additional significant feature noted in this study is the relevance of context within sentiment analysis. In spite of the fact that the Bag of Words technique does not consider the order of words, it was still possible for the models to identify some meaningful features. Nevertheless, it was also mentioned that more sophisticated methodologies would be required in the future to deal with contexts and linguistic constructs.

Key Points from the study :

- Introduced the use of machine learning algorithms for sentiment classification instead of relying on manual rule-based systems
- Demonstrated that models like Naive Bayes, Maximum Entropy, and SVM can effectively learn patterns from labeled data
- Showed that machine learning approaches provide significantly higher accuracy compared to traditional methods
- Used Bag of Words as a simple and effective feature extraction technique for converting text into numerical form

- Highlighted the importance of training data in improving model performance
- Identified limitations of early methods, especially in handling context and complex sentences
- Laid the foundation for future research in sentiment analysis, including advanced NLP and deep learning techniques
- Proved that automated sentiment analysis can be applied to real-world data such as customer reviews

2.2 Go, A. et al. - "Twitter Sentiment Classification using Distant Supervision"

Convenient Name: Go et al. (2009) – Twitter Sentiment Analysis
Source: Stanford University Research

This research is considered to be among the most significant papers in the area of sentiment analysis based on real-world social media content. In this study, the authors concentrated on the process of sentiment analysis of tweets, which are very brief text messages created by users. The novelty of this research lies in the use of distant supervision for automatic creation of labels.

In this study, the researchers used many tweets, with emoticons such as denoting positive emotion and denoting negative emotion. Those tweets that had a positive emoticon were tagged positive, while those that had a negative emoticon were tagged negative. This made the creation of a massive dataset possible with no need for manual work, thus speeding up the training process.

The dataset was then analyzed using algorithms like Naive Bayes, Maximum Entropy, and Support Vector Machines (SVM) to determine the sentiment associated with each tweet. The outcome indicated that these algorithms worked well even with the noisy text data used. This shows that machine learning algorithms work well with text data, even when the data has slang, abbreviations, and grammatical errors.

The second valuable point is the consideration of big data in this paper. In comparison to previous studies, which were based on a small number of samples, this study revealed that a huge volume of data is able to enhance sentiment analysis algorithms' performance. As more training data is provided to the algorithm, the more efficient it works. There are also some difficulties associated with analysis of social media posts. Tweets are often brief and have various misspellings and informal vocabulary. Moreover, they may include hashtags and other characters. Still, the accuracy was rather good.

Key Points from the Study :

- Introduced the concept of distant supervision for automatic data labeling, reducing the need for manual annotation
- Used emoticons as sentiment indicators to create a large labeled dataset efficiently
- Demonstrated that machine learning models can handle noisy and informal real-world data
- Showed that large-scale datasets improve the accuracy and performance of sentiment analysis models
- Applied algorithms like Naive Bayes, Maximum Entropy, and SVM for classification tasks
- Highlighted challenges such as slang, abbreviations, and short text length in tweets
- Proved the scalability of sentiment analysis systems using big data
- Provided a foundation for analyzing user-generated content in various domains
- Techniques are highly applicable to e-commerce review analysis and customer feedback systems

2.3 Liu, B. — "Sentiment Analysis and Opinion Mining"

Convenient Name: Liu (2012) – Opinion Mining
Source: Morgan & Claypool Publishers

This book by Bing Liu is regarded as one of the most complete and useful references on sentiment analysis and opinion mining. This book explains the techniques and approaches that are used for analyzing opinions based on textual data. The book explains everything ranging from the fundamental to the more advanced aspects of sentiment analysis, hence making it a very useful resource for this purpose.

One of the key contributions of this work is the explanation of different levels of sentiment analysis. Liu describes three main levels:

- Document-level analysis, where the overall sentiment of an entire review or document is classified as positive, negative, or neutral.
- Sentence-level analysis, where each individual sentence is analyzed separately to determine its sentiment.
- Aspect-level (feature-level) analysis, which is the most detailed approach. It identifies specific features or aspects of a product (such as quality, price, delivery, etc.) and determines the sentiment for each aspect.

The importance of this technique in practical application domains such as e-commerce is highlighted in the research, as each review can simultaneously contain both positive and negative sentiments concerning various aspects of the subject matter. For instance, a consumer may be satisfied with the quality of the product but unsatisfied with the shipping service.

The other significant contribution made by this study relates to opinion mining methodology, which entails the identification and extraction of subjective text data such as opinions, feelings, and attitudes. This process involves the detection of opinion terms, opinion holders, and opinion targets.

Sentiment analysis poses various difficulties, too. One of the main difficulties is sarcasm detection, when the real message of a sentence is completely different from what the sentence means literally, e.g., “This is the best product ever… not!” is a negative message despite what the sentence says. Mixed sentiment analysis, which refers to the process of analyzing the mixture of negative and positive messages within one sentence/review, is another difficulty. Finally, the complexity of language is also a problem.

The concept of domain dependency is also mentioned in the book. In other words, words that have positive sentiments in one domain could also have negative sentiments in another domain. For instance, a word such as "cheap" could have a positive sentiment when associated with pricing, whereas it could be negative when associated with product quality. This means that sentiment analysis models should be designed with particular consideration of the domain. This literature has largely influenced modern approaches to sentiment analysis. Several advanced approaches and models for sentiment analysis have emerged from this theory.

Key Points from the Study:

- Provides a comprehensive overview of sentiment analysis and opinion mining techniques
- Introduces different levels of sentiment analysis: document-level, sentence-level, and aspect-level
- Explains the importance of aspect-based sentiment analysis for detailed and meaningful insights
- Describes methods for extracting opinions, opinion holders, and opinion targets from text
- Highlights real-world challenges such as sarcasm detection, mixed sentiments, and language complexity
- Discusses domain dependency and how sentiment can vary across different contexts
- Emphasizes the need for advanced NLP techniques to improve accuracy
- Serves as a foundational reference for researchers and developers in sentiment analysis

2.4 Kim, Y. — "Convolutional Neural Networks for Sentence Classification"

Convenient Name: Kim (2014) – CNN for Text Classification
Source: EMNLP Conference

In this research paper, CNN was utilized for sentiment analysis purposes in text classification. Before this paper, the majority of approaches to sentiment analysis were based on conventional machine learning models using Bag of Words or TF-IDF as the feature extraction mechanism. This approach required intensive labor to choose and design features.

According to Kim's experiment, the ability of a CNN to automatically detect useful patterns from text without the need for manual feature extraction was demonstrated. In the process, the model takes words represented numerically into a network through a word embedding process and then uses convolutions to detect patterns within the sentence.

An important strength of CNNs is the ability to learn local features present in the text, including commonly occurring word sequences. Words such as "very good," "not satisfied," and "highly recommended" may easily be learned by the model. Through the use of multiple filters, the network is able to learn various features in the input text data.

Experiments were performed using several data sets in this study, and it was found that CNN models outperform conventional machine learning algorithms in terms of accuracy. This clearly indicates that deep learning models perform better when it comes to comprehending complex patterns within language and thereby improving sentiment prediction. Another crucial feature of this study is that it has made use of pre-trained word embeddings such as Word2Vec. Such embeddings assist the algorithm to comprehend the meanings of words and their relationship.

Limitations of CNN were mentioned as well in the paper. CNN is computationally expensive, and it requires substantial amounts of training data. Besides, it is more complicated than conventional models and can be difficult to apply and optimize. However, CNNs have been widely used for text classification owing to their superior accuracy rates despite the difficulties mentioned above. In addition, CNN has made a great contribution to the use of deep learning in sentiment analysis and encouraged many researchers to use other models like RNNs and LSTMs.

Key Points from the Study:

- Introduced the use of deep learning (CNN) for sentiment analysis and text classification
- Demonstrated that CNNs can automatically learn features from raw text data without manual feature engineering
- Captured important patterns and phrases in sentences using convolution filters

- Achieved higher accuracy compared to traditional machine learning models
- Utilized word embeddings to understand semantic relationships between words
- Improved the ability to handle complex and large-scale datasets
- Highlighted advantages in detecting local text patterns and meaningful phrases
- Identified challenges such as high computational cost and need for large training data
- Inspired further research in advanced deep learning models for NLP tasks

2.5 Devlin, J. et al. — "BERT: Pre-training of Deep Bidirectional Transformers"

Convenient Name: Devlin et al. (2018) – BERT Model
Source: Google AI Research

In the paper "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", BERT (Bidirectional Encoder Representations from Transformers) was presented. This model is considered one of the most advanced models in the domain of NLP. Prior to BERT, the majority of models used a one-sided approach where the sentences were processed either from left to right or vice versa. The limitation of this technique was that there was no complete understanding of the sentences' context. With BERT, the problem was solved through bidirectional processing.

One of the features that makes BERT unique is its application of the transformer model, which makes use of an attention function. It enables the model to identify and understand relationships between significant words in a sentence, thus allowing BERT to analyze complex linguistic information that was previously difficult for other models to discern.

BERT also goes through pre-training with a huge amount of text data and fine-tuning based on different tasks like sentiment analysis, question answering, and text classification. This makes the model versatile and very efficient. Take, for instance, sentiment analysis. In this case, BERT is capable of analyzing complicated sentences and even understanding sarcasm.

The experiment showed that BERT outperformed other language models significantly in various NLP tests and achieved the state-of-the-art results for its time. It became one of the most commonly used models for academic research and practical applications. An important characteristic of BERT is its ability to cope with cases of polysemy, when there is more than one meaning for a certain word. The term bank, for instance, could be interpreted either as a financial organization or the riverbank.

However, despite its benefits, BERT has some disadvantages. Its application demands great computational power, large memory, and long periods for training, which makes the tool quite demanding. The fine-tuning process is also quite sophisticated, requiring proper adjustment of parameters in order to obtain maximum results. This study was very important for NLP as well as for improving the performance of sentiment analysis tools. In addition, it was an impetus for creating other advanced models in artificial intelligence.

Key Points from the Study:

- Introduces a transformer-based architecture, which is highly effective for advanced Natural Language Processing (NLP) tasks.
- Uses bidirectional context understanding, allowing the model to read and interpret text from both left and right directions for better meaning extraction.
- Applies attention mechanisms to identify and focus on the most important words in a sentence, improving overall understanding.
- Achieves state-of-the-art performance in sentiment analysis and various other NLP tasks, outperforming traditional models.
- Capable of understanding complex language structures, sarcasm, and contextual meaning, which are challenging for earlier approaches.
- Pre-trained on large-scale datasets and can be fine-tuned for specific applications, making it highly flexible and powerful.
- Handles multiple meanings of words (polysemy) effectively by analyzing the context in which they are used.
- Widely used in real-world applications such as chatbots, search engines, and recommendation systems.
- Requires high computational resources, memory, and processing time, which can be a limitation for small-scale implementations.

- More complex to implement and optimize compared to traditional machine learning models.

2.6 Ghadekar et al. — "Sentiment Analysis on E-commerce Reviews using Machine Learning"

Convenient Name: Ghadekar et al. (2023) – E-commerce Sentiment Analysis **Source:** Research Publication

This research is specifically aimed at studying customer reviews on e-commerce websites and is therefore highly pertinent to our proposed research project. The aim of this research is to classify customer reviews according to positive, negative or neutral sentiment through the use of machine learning algorithms.

This system starts with data acquisition, where the customer review data is collected from e-commerce websites. The presence of noise in the raw data in the form of punctuation, special characters, or unnecessary words makes data preprocessing essential for accurate results. Data preprocessing can be done using techniques like tokenization, stop word removal, and stemming, among others.

The preprocessing stage is followed by the implementation of TF-IDF (Term Frequency-Inverse Document Frequency) which is a method of feature extraction in the paper. TF-IDF represents text in numerical terms by highlighting key words in a review. Key words are those that occur many times in a particular review but do not occur often in other reviews. Such an approach is useful because the model can focus on the actual contents and avoid words that may not be important. Logistic Regression, a machine learning method used commonly in text classification tasks, is implemented in the analysis.

In addition, the performance of the model has been assessed through measures like accuracy, precision, recall, and F1 score. Such performance tests ensure that the developed model is accurate and provides consistency. From the results, it has been observed that machine learning methods have proven to be accurate enough and hence appropriate for practical purposes.

Another significant contribution made by this research relates to its practical implications. This study shows the applicability of sentiment analysis in practical business settings, contrary to theoretical research studies that are not based on any practical use case. Using customer reviews, organizations can find out what issues prevail among their customers, and based on their likes and preferences, they can improve their products or services accordingly. The other significant contribution relates to scalability because the amount of data generated in e-commerce websites is huge on a daily basis.

Some limitations have been highlighted as well within this particular piece of research. For example, the system in question has difficulties dealing with complex sentences, sarcastic remarks, and mixed emotions, which cannot be solved by the use of the described techniques and call for more sophisticated tools such as deep learning. Nevertheless, this research offers a solid basis for developing useful models for analyzing sentiments. It is highly pertinent to the suggested project in that similar techniques are utilized.

Key Points from the Study:

- Focuses on real-world e-commerce customer review data, making the study highly practical and relevant for business applications.
- Uses data preprocessing techniques such as removing stop words, punctuation, and noise to clean and prepare text for accurate analysis.
- Applies TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual data into meaningful numerical features for model processing.
- Uses Logistic Regression, which is an efficient and reliable machine learning algorithm for sentiment classification tasks.
- Evaluates model performance using standard metrics such as accuracy, precision, recall, and F1-score to ensure reliability and effectiveness.
- Demonstrates the practical implementation of sentiment analysis in real-world business scenarios, especially in e-commerce platforms.
- Helps businesses understand customer opinions, identify common issues, and improve decision-making based on data-driven insights.
- Supports scalability, allowing the system to handle large volumes of customer reviews efficiently.

Chapter 3

METHODOLOGY

The approach used in building the AI-Powered Customer Sentiment Analysis tool in e-commerce is through integration of different software tools, programming approaches, and machine learning techniques. Each particular approach has an individual role to ensure that customer reviews are processed in an efficient manner and that insights are drawn from textual data. The present chapter presents the description of how each of the major software tools, Python, Machine Learning, NLP, TF-IDF, contributes to the development of the tool.

3.1 System Workflow

Workflow of the System for the AI-Powered Customer Sentiment Analysis project can be described as an orderly process in which raw reviews provided by customers are processed in order to provide sentiment analysis insights. Workflow of the system helps to understand the flow of data in the system from the very beginning (i.e. collection of raw data) until the end (providing of final insights).

These include processes such as data preprocessing, feature extraction, modeling, evaluation, and prediction. Each of the processes described above plays an important role in enhancing the performance of the system by producing accurate analyses. Using an effective workflow for sentiment analysis involves converting unstructured text into numeric data, applying machine learning algorithms, and producing accurate classifications. Notably, using a systematic workflow does not only enhance the performance of the system but also ensures that its implementation and maintenance become simple and straightforward. Moreover, using the system workflow allows the complete process of analyzing sentiments of customers to be automated using minimum human intervention. The workflow allows the system to analyze customer sentiments in real-time, extract opinions of consumers within a short period, and produce various visualizations such as graphics, graphs, and charts to

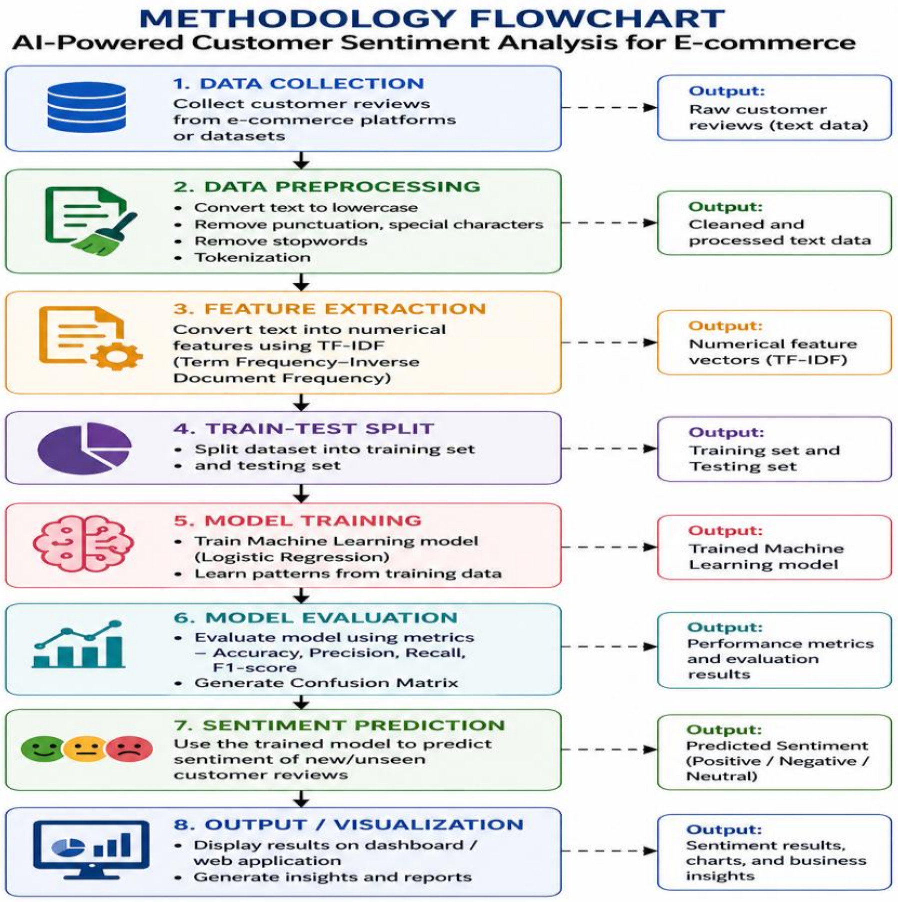


Fig 3.1 : Flowchart

1. Data Collection : Data Collection stage involves collecting the customer reviews on e-commerce sites or from databases. Customer reviews can take various forms including opinions, feedbacks, and ratings, among others. The data obtained is in raw text form and is not fit for any form of analysis at this point; hence, it requires processing.

2. Data Preprocessing : Once the data collection is complete, the next process that follows is the Data Preprocessing process. It involves the cleaning and preparation of the collected data for analysis. Some of the tasks involved include converting the data into lowercase letters, getting rid of punctuations, removing special characters and stop words such as 'is', 'the', and 'and', among others, and tokenization.

3. Feature Extraction : After the data has been cleansed, the next phase is the extraction of features. The text data is transformed to numbers through the use of the TF-IDF algorithm. As we know, machine learning algorithms cannot understand the text data directly, and thus the words have to be translated to numbers according to their significance in the text.

4. Train-Test Split : Here, the dataset is partitioned into two categories; training data and testing data. The former data is utilized to develop the machine learning algorithm whereas the latter is used to test its effectiveness. This way, the machine learning model can be tested against unseen data rather than memorizing the dataset.

5. Model Training : Once the data split is done, the Model Training stage is initiated. Here, machine learning techniques like Logistic Regression can be applied. This model will be trained using the training data set by recognizing the associations between words and sentiments.

6. Model Evaluation : After training the model, it is tested for assessing its performance. Different measures, like accuracy, precision, recall, and F1 score, are calculated to evaluate the performance of the machine learning model. The confusion matrix is used to identify the accuracy and inaccuracy in the prediction of the model.

7. Sentiment Prediction : Once the model is evaluated, the next stage in the process is that of Sentiment Prediction. Here, inputs of previously unseen data in the form of customer feedback are presented to the system, which will then predict whether the sentiment is positive, negative, or neutral.

8. Output / Visualization : Results are shown at the Output/Visualization phase. The prediction is displayed in the form of a dashboard or through a web-based application interface. Other visualizations like charts, graphs, and even reports can be developed in order for companies to gauge customer sentiments.

3.2 Frontend Technologies

3.2.1 HTML

HTML is an abbreviation for Hyper Text Markup Language. HTML is considered to be the core or fundamental language for building web pages. The job of HTML is to organize the web pages with the help of various components like heading, paragraphs, images, hyperlinks, forms, and many more items that appear on the web page. However, HTML doesn't deal with any logic or calculation.

HTML works using tags, which are written inside angle brackets (<>). These tags tell the browser how to display the content. For example, <h1> is used for headings, <p> is used for paragraphs, and <img> is used to display images. Most HTML tags have an opening tag and a closing tag, and the content is written between them. This makes it easy to structure and format information on a webpage.

HTML will be employed in this project to build the user interface (UI), through which the user will be able to provide reviews about the customers and obtain the result in terms of sentiment prediction. HTML is useful in building interfaces with input fields, buttons, and outputs such as positive, negative, and neutral sentiments.

HTML is easy to master, compatible with all web browsers, and serves as the base of web development. It is impossible to design a website without the use of HTML; thus, it is critical in developing a web system.

3.2.2 CSS

The abbreviation CSS refers to Cascading Style Sheets. This tool is utilized for styling and laying out HTML web pages, which means CSS gives control over the visual representation of pages that are developed in HTML. While HTML serves for building a framework of the page, CSS makes the page more appealing and convenient for users. By using CSS, web developers can set various styles for HTML elements, including colors, font size and type, positioning, borders, backgrounds, margins, padding, and others. CSS helps to make the website more attractive and comfortable to use for its visitors. There are several methods of applying CSS, namely, inline, internal, and external CSS.

For this project, the role of CSS is utilized to create an interface for the sentiment analysis system. This language is useful in styling form elements through which review inputs are taken from the user, buttons for submission purposes, and parts of the output displaying sentiments such as positive, negative, and neutral. The CSS language contributes toward keeping the website structure clean and good-looking.

The use of CSS is also effective in developing a responsive design where the application adjusts automatically on screens of different sizes, including mobile devices and desktop computers. This makes the design much more professional and easy to use by everyone. The CSS design language plays a crucial role in improving the look of the web application and increasing its interactivity.

3.2.3 JavaScript

The role of JavaScript is to increase the interactivity and responsiveness of your website. If you have undertaken an eCommerce project, then using JavaScript enables you to interact with the web pages to add/remove products, search for products, or filter categories, all without having to refresh the web page each time.

Another application of JavaScript is its use in validating forms. When entering details like the name, email, or address in the checkout process, the script validates the data before forwarding it. This prevents any mistakes from being made. JavaScript is used in updating content dynamically. The script can be used to update the price of the item, amount of money left in your shopping cart, or even stock quantity.

The other significant use of the JavaScript programming language is event handling. This is used when performing functions such as clicking a button, mousing over an image, scrolling through the page, or picking up something. For instance, when you click "Add to Cart," it will automatically change the number of items added to your shopping basket. JavaScript can help improve the UI/UX of your website by adding things like animations, pop-ups, sliders, and drop-down menus.

3.3 Backend Technologies

3.3.1 Python

Python is the main programming language employed for developing the backend architecture and integrating machine learning models as well as for performing other data processing tasks. The choice is made in favor of Python because it is a rather simple and readable programming language with a large community of developers and numerous libraries for data science, natural language processing, and web technologies.

Python has a key part in designing the entire workflow for the project, ranging from user inputs all the way through to sentiment predictions. This is because of the fact that Python can be used to implement machine learning techniques, such as Logistic Regression, Decision Trees, and Random Forests. All of these techniques will be used to sort out the reviews made by the customers.

In this case, Python is used as the engine of the backend process. Once an individual submits their review from the online store, the system gets the data input via the APIs created using the framework Flask, cleans up the text by getting rid of all the stop words and punctuation marks, and tokenizes the data. Thereafter, the cleaned text is converted into numeric values using the TF-IDF vectorizer and is passed through the machine learning model of sentiment analysis before sending results to the front end.[9]

The use of several Python libraries makes this application efficient and scalable. For instance, the Pandas library is employed for handling datasets, Scikit-learn is applied to train and test the machine learning model, NLTK is employed in natural language processing, while Flask is used for developing APIs on the backend.

The program also allows for future enhancements and scalability of the application. Features such as deep learning sentiments analysis through the use of LSTM or BERT algorithms, live analytics dashboard features, and even integration with cloud systems such as AWS and MongoDB Atlas can easily be incorporated. In doing so, it is easy to make the software more advanced and intelligent.

3.3.2 Flask

Flask is a minimalist web application framework written in Python that is applied in your AI-based e-commerce platform where you implement the backend of your site. Flask plays an important role in connecting the backend (Python programming environment) and the front end (the actual website).

The main role of Flask is in developing APIs (application interfaces), which connect the user interface with the backend of the program. For instance, whenever a customer posts a comment or a feedback on your e-commerce site, Flask passes that data into the machine learning environment.

Flask works as follows:

• It receives customer reviews or input data from the frontend (HTML/CSS/JS).
• It routes the request to the correct Python function using API endpoints.
• It sends the review text to the preprocessing and ML model pipeline.
• It returns the predicted sentiment (positive, negative, or neutral) back to the frontend.
• It helps display results dynamically without reloading the page.

The other advantage of using Flask is that it will make your work more interactive and real-time. Rather than running your code in Python scripts manually, Flask enables your site to interact with the model you have created through web requests. Besides, Flask is light and can be easily integrated into your project since it has all the tools needed to implement a project at your level.

Also, Flask is easy to integrate into libraries such as Scikit-learn, Pandas, and NLTK that will be used in your sentiment analysis project. Through Flask, you can create an efficient connection between the data preparation process, predictions using the model, and data display on the frontend.

3.2.3 MongoDB

Introduction of MongoDB Atlas: MongoDB Atlas is the cloud database provided by MongoDB that is used in your customer sentiment analysis project powered by artificial intelligence. This database is used as the main database system in which all data of your project, including customer reviews, user data, and sentiment results, are saved.

Application of MongoDB Atlas in Our Project: In our project, MongoDB Atlas plays an important role in saving the data of customer reviews. When a user provides feedback on the product purchased from the website, the data is then passed from the frontend to the backend (Flask) and then it is saved in MongoDB Atlas in the form of documents.

The database architecture used by MongoDB Atlas is NoSQL, which implies that data storage in a form similar to JSON (BSON). This is highly beneficial for your case study since the customers' reviews as well as sentiment analysis output will be considered as unstructured data and might differ in size and format. As far as our system operation cycle goes, MongoDB Atlas will play an essential role in retrieving and analyzing collected data. In other words, your backend will be able to retrieve stored reviews from the database in order to conduct sentiment analysis or train models using that data.

Real-time data updating in MongoDB Atlas is another important feature for your e-commerce project. Whenever there are new reviews in the database, it will be updated immediately without any further action from you, and your sentiment analysis application will work on the latest data.[12]

Scalability and Cloud Storage is yet another benefit offered by MongoDB Atlas. As it is a cloud-based system, you will not need to rely on any sort of local storage for your application, thus allowing you to store plenty of data when the e-commerce application grows in size. The system offers high availability as well as data security. MongoDB Atlas serves as the core data storage element of your application.

3.4 Machine Leaning Technologies

Machine Learning is an area within Artificial Intelligence (AI) that allows computer systems to learn through data without having to be programmed for specific operations. Machine learning is based on the analysis of data patterns rather than predetermined sets of rules.[8]

3.4.1 TF-IDF Vectorization

What is the TF-IDF vectorization? This technique is one of the features that extract methods which helps in turning the natural language processing information into numeric values. Why? Because algorithms do not deal with texts, so TF-IDF becomes essential for converting customer reviews into numeric values.

TF-IDF algorithm is employed for sentiment analysis in customer reviews in ecommerce. The algorithm uses weights to determine the significance of a word. This means that words that have a high frequency in the particular review but low frequency in all other reviews will be given more significance. Words like good, excellent, bad, and poor can be captured using TF-IDF technique.

TF-IDF comprises two key elements – Term Frequency (TF) that assesses the frequency of appearance of words in one particular document and Inverse Document Frequency (IDF) that examines their rarity or ubiquity in all documents. This enables the TF-IDF to provide lesser importance to the frequently used words such as "the" or "is" and emphasizes more on meaning-filled words.

In this sentiment analysis system, TF-IDF plays a vital role in transforming the cleaned customer reviews into numerical vectors for Logistic Regression to process. This increases the accuracy of classification of the sentiments as the algorithm will focus on the relevant words in order to understand users' views. In turn, the system will accurately classify the review based on its sentiment into positive, negative, and neutral. Without using TF-IDF, the system would not be able to analyze sentiments in reviews.

3.4.2 Classification Algorithms

- **Logistic Regression :** Logistic regression is a form of supervised learning algorithm which is used for classification problems where the outcome can be in terms of sentiment being either positive or negative. Your project uses this logistic regression algorithm to classify the review sentiments into positive (1) or negative (0). The fact that the word "regression" is included in the term doesn't mean anything.

  The functioning of this model relies on computing the probability that an input data belongs to a certain category through the application of a sigmoid function. The range of the output value is from 0 to 1. If the probability exceeds a certain threshold (typically 0.5), then the input data is regarded as positive; otherwise, the input data is labeled as negative. The model learns the association between input features (TF-IDF value of words) and output labels from the training set. Logistic Regression is popular for text categorization tasks such as sentiment analysis due to its efficiency and effectiveness, especially with big data sets. Your project proved that it is the most accurate model, hence chosen as the final one.

- **Naive Bayes :** Another supervised machine learning algorithm is Naïve Bayes used mainly for performing classification. Like most machine learning algorithms, it uses probability, and Bayes' Theorem is used to perform the calculation of the probability of the classes based on some attributes.

  For the purpose of the project, the algorithm, known as Naive Bayes, determines if a review is positive or negative based on the likelihood of certain words to appear in either category of reviews. The assumption here is that each word in a sentence does not depend on any other word in a sentence, hence the name "naive". The algorithm works really well despite the false assumptions made.

  Naive Bayes is a fast algorithm that is simple to code and performs excellently when applied to smaller data sets. Nevertheless, Naive Bayes can be less accurate than logistic regression for some specific data sets. Naive Bayes was applied during your project, but since logistic regression worked better, it was selected. [6]

- **Support Vector Machine (SVM) :** Support Vector Machines (SVMs) are algorithms that fall under the category of supervised machine learning algorithms and can be used for classification and regression purposes. In our AI-Powered Customer Sentiment Analysis for an E-Commerce Website project, SVM will help us classify the sentiments expressed by customers, which may include Positive, Neutral, and Negative sentiments.

Our project involves customers writing reviews for items including shoes, clothes, gadgets, or any other items. The reviews are in text form and cannot be interpreted directly through machine learning algorithms. As such, before applying SVM, the text data needs to undergo some preprocessing in which all unnecessary symbols are removed and text converted to lowercase while at the same time being changed to numbers using techniques such as TF-IDF.

The primary working principle of the SVM technique includes determining a separation boundary known as a hyperplane between various classes of sentiments. The algorithm focuses on determining an optimal separation boundary for which there is maximum distance between the classes. This means that while determining reviews such as “Amazing quality shoes and fast delivery” as being positive, while "Very poor quality and waste of money" are negative sentiments.

Another advantage of SVM in our project is its ability to provide better classification accuracy and reduce misclassification errors. Compared to some traditional algorithms, SVM [5] often gives strong results for text classification problems because it focuses on finding the optimal decision boundary. By using SVM in the AI-powered sentiment analysis system, the website can automatically analyze customer opinions, calculate sentiment scores, generate ratings, and help users make better purchasing decisions. This improves the overall customer experience and provides valuable insights for both customers and e-commerce businesses.

3.5 Deep Learning Models

- **Convolutional Neural Network (CNN) :** Convolutional Neural Network (CNN) is a deep learning architecture primarily used in image processing; however, it performs well in text classification.

  In this study, CNN was applied in extracting important local patterns in text, such as phrases and word sequences that affect sentiment. CNN uses convolutional filters on word embeddings in order to derive useful information from reviews. Even though CNN gave satisfactory results, it took longer processing and training time.[3]

- **Recurrent Neural Network (RNN)** : RNN is designed to work on sequences like language data. RNN analyzes words in sequential order and holds the memory of previously fed input data, which makes RNN capable of understanding the context.

  RNN is applied on review sequence analysis to determine the sentimental flow. But this model was having longer training time, along with vanishing gradient problems.

- **Long Short-Term Memory (LSTM)** : LSTM stands for Long Short Term Memory which is a very sophisticated neural network designed specifically for dealing with text data. In your case, where you are implementing your project "AI-Powered Customer Sentiment Analysis for E-Commerce Website", your text data (reviews) are in the form of sentences and the order of words matters. Machine learning models usually have difficulties in comprehending long sentences and the overall context. LSTM helps by allowing them to remember the useful part of the sentences and forgetting the rest.

  A model that uses LSTM can be developed using 100-300 neurons in the hidden layer. Its basic structure consists of four layers, which include the Embedding Layer, the LSTM Layer, the Dropout Layer, and the Dense Layer. The purpose of the embedding layer is to transform words into numerical vector representations; the LSTM layer captures patterns from customers' opinions; the dropout layer helps minimize overfitting; and the dense layer predicts sentiments.

  The reason behind the better accuracy of LSTM over traditional RNN models is its capability to retain long-term information. The algorithm can be used for sentiment analysis since customer feedback typically consists of lengthy sentences. On the other hand, training LSTM requires more time and resources than traditional machine learning algorithms like LR and SVM. [4]

- **Bidirectional Long Short-Term Memory (BiLSTM) :** BiLSTM is the enhanced form of LSTM. In general, LSTM scans the text data in only one direction, mostly from left to right. However, in BiLSTM, scanning happens in both directions, which means that it scans text data from left to right as well as from right to left simultaneously.

BiLSTM helps to scan texts more effectively. If we analyze sentiment analysis project data using BiLSTM, it would be easy for us to understand the context in reviews which depends on the upcoming words as well. Let us take an example: "I thought these shoes were great, but then realized serious quality problems with them."

The structure of BiLSTM is composed of Forward LSTM, Backward LSTM, and Dense layer. Just like LSTM, the number of neurons employed is in the range of 100 to 300. Because of learning information in two directions, more profound relationships can be learned.

- **Bidirectional Encoder Representations from Transformers (BERT):** BERT stands among the top NLP models designed to make sense of language text. While traditional machine learning and even LSTM models depend on other approaches to understand the relation between words, BERT utilizes the Transformer model with the Attention method.

  In the process of analyzing customer sentiment using an AI-based model, BERT is capable of comprehending the complete meaning of the review because of its ability to consider the context of the entire sentence. For instance, in case a consumer states that "the shoes are light but the durability aspect let me down," BERT [2] will be able to recognize how all words interrelate and make accurate predictions about the sentiment expressed in the statement. BERT consists of millions of parameters and various layers like embedding, transformer, and attention layers.

  The accuracy of BERT is relatively very high when compared to other algorithms in terms of sentiment analysis. This method can give outstanding performance since it has the ability to comprehend human language context. But BERT requires highly powerful computing machines like GPUs and needs more memory and processing time. For academic purposes, many scholars make use of BERT models that have already been trained.

Model	Type	Accuracy
Logistic Regression	ML	Highest
Naive Bayes	ML	Moderate
SVM	ML	High
CNN	Deep Learning	Good
RNN	Deep Learning	Good
LSTM	Deep Learning	High
BiLSTM	Deep Learning	Very High
BERT	Deep Learning	Highest

3.6 Libraries and Tools Used

1. Data Handling

- **Pandas :** Pandas is employed to work with structured data, helping to read datasets (CSV files), clean data, and organize review data in tabular form.

- **Numpy :** NumPy is utilized to perform calculations involving numbers. NumPy helps perform quick mathematics involving arrays and is thus beneficial while dealing with huge datasets.

2. Web Scraping

- **Requests :** The Requests library is employed to make HTTP requests to any website. This library helps in retrieving webpages from online shopping sites so that reviews can be extracted.

- **BeautifulSoup :** The BeautifulSoup library is used to extract information from HTML content. This library helps extract useful information from webpages, such as product reviews.[10]

- **Selenium :** Selenium is an open-source library that helps perform actions like clicking buttons, filling up forms, searching data, and scraping websites using automated control over web browsers.[11]

3. Natural Language Processing (NLP)

- **NLTK :** NLTK (Natural Language Toolkit) is used for performing operations such as stopword removal, text tokenization, and preparation of reviews for modeling purposes.

- **scikit-learn :** This is an ML library used for modeling tasks. It provides functionalities for TF-IDF vectorization, model training, predictions, and evaluation.

4. Model Saving

- **joblib / pickle :** These libraries are used to save trained machine learning models. They allow the model to be stored and reused later without retraining.

Chapter 4

SYSYEM ANALYSIS

E-commerce websites receive many reviews by consumers who express their experience with the product. These reviews are critical to both buyers and sellers as they determine purchases and improve the products. But due to high numbers, analyzing such a huge amount of information is challenging to both consumers and firms manually. Going through all the comments will be tedious and inefficient.

This problem could be solved with the use of an automated AI-driven sentiment analysis program which analyzes customer review texts using advanced machine learning techniques and natural language processing tools to make sense out of them. Instead of reading through each review individually, this software automatically analyzes textual data for sentiment and provides valuable insight such as rating and overall product assessment for better understanding of its quality.

Web scraping techniques are utilized for retrieving reviews on e-commerce websites. Data preprocessing is done next, where the text is converted into numerical representation using TF-IDF method before being fed to either logistic regression or naive bayes classifier models trained beforehand. Finally, the results are presented in a convenient web application created using Flask framework along with other HTML/CSS/JS technologies.

The benefits of this model include improved decision-making by customers, information on product feedback for sellers, and elimination of human labor. The system offers more accurate and faster review analysis than the conventional methods.

4.1 Functional Requirements:

Functional requirements explain all the processes that should be done by the system in order to reach the goal set for it. Functional requirements for the AI-driven customer sentiment analysis system include processing the input from the user, analyzing data, generating results, and displaying results.

1. User Input and Website Interaction

Users should have the ability to enter either the product URL or provide the review text manually to use this software. The user entry must undergo validation to check whether it conforms to the proper format. In case the URL entered by the user is incorrect or missing altogether, an error message should be displayed. It is important to note that the application is meant for e-commerce product URLs.

2. Web Scraping of Reviews

The system will make use of web scraping techniques for collecting customer reviews from online shopping websites. Requests and BeautifulSoup libraries will be used for fetching pages and extracting information respectively. The system is expected to gather reviews effectively from one product page. This should include handling dynamically generated pages or structured data on the web and extracting only the necessary parts of the reviews.

3. Data Preprocessing

The system will be required to clean and preprocess the review text. This process includes the transformation of the text to lowercase, elimination of all special characters, punctuations, and other unnecessary characters. Words such as "is," "the," "and" should also be eliminated to ensure quality processing. The preprocessed data will then be transformed into a standard format to enable machine learning.

4. Feature Extraction

The system will convert the data from text to numeric form using the TF-IDF technique. This will facilitate the representation of words that are important depending on the relevance of those words within the database. The system produces structured vectors of numbers which machine learning models can use for classification purposes.

5. Sentiment Classification

Customer feedback is to be classified under either positive or negative sentiment classes. This will be done using algorithms like logistic regression or naive bayes after training. In making this classification, it takes into account patterns within the text of the review in predicting whether it is either a positive or negative one.

6. Rating Generation

The system shall provide product ratings depending upon the sentiment expressed in reviews that have been collected. After classifying them, the system finds out the number of positive and negative reviews and converts it into a rating scale ranging from 1 to 5. The greater the number of positive reviews, the higher the rating will be.

7. Result Display

The system shall use Flask, HTML, CSS, and JavaScript to present the sentiments' analysis through a web interface. It presents the end sentiment, ratings, and a summary of the reviews provided. The web interface design will ensure that the presentation of the analysis is done in a way that can be easily interpreted by everyone.

8. Model Training and Prediction

The system shall utilize a dataset consisting of the reviews' texts labeled based on their sentiments to build machine learning models. The learned model shall then be used for classification purposes on new unseen reviews. The prediction process will be very fast.

9. Data Storage

All processed review data, training data, and prediction outputs should be saved in the system to facilitate future use. This is important to ensure that there will be no need to process any data that can be used again. This facilitates system improvement and future analysis of the data if need be.

10. Error Handling

The system should have proper mechanisms to deal with any error during execution. In case the wrong URL has been entered or there are no reviews in that URL, the system should display appropriate error messages.

11. Security and Access Control

The system will also guarantee that any data received from users or processed internally is handled securely. Access to the different components of the system by any unauthorized person must not be allowed.

12. Model Saving and Loading

The system should have the ability to save trained machine learning models using joblib or pickle libraries. This would help the system in saving the trained model, without the need to train the model each time. It saves a lot of time for prediction purposes.

13. System Logout / Session Handling

The system should have an ability to handle user sessions in the web application. The system should give an opportunity for users to logout once their work is completed. The system should automatically terminate a session after some time.

4.2 Non-Functional Requirements:

The non-functional requirement is about how good the system is at performing certain operations. It involves the quality features of the system like performance, accuracy, reliability, security, and usability in regard to e-commerce review sentiment analysis.

1. Performance

- The system processes reviews quickly using efficient ML and NLP techniques.
- Web scraping, preprocessing, and prediction are completed in very less time.
- TF-IDF and classification models work efficiently even with large datasets.
- The system supports multiple requests without slowing down.

2. Accuracy

- The system provides high accuracy in sentiment prediction.
- It correctly classifies reviews as positive or negative.
- Proper preprocessing improves prediction quality.
- Output closely matches real customer opinions.

3. Reliability

- The system works consistently without frequent errors or crashes.
- It handles large review data without failure.
- Predictions remain stable for similar inputs.
- Trained models and data remain safe during execution.

4. Usability

- The system is simple and easy for all users.
- Non-technical users can operate it easily.

- The interface is clean and user-friendly.
- Results are displayed in a clear and understandable format.

5. Scalability

- The system can handle increasing number of users and reviews.
- It supports multiple product analyses at the same time.
- Performance remains stable even with large data.
- Future upgrades like new models can be added easily.

6. Security

- The system protects user data from unauthorized access.
- Backend services and APIs are secured properly.
- Only valid users can access system features.
- Data integrity is maintained throughout processing.

7. Privacy

- User inputs like URLs and reviews are kept confidential.
- Data is not shared with unauthorized users.
- All information is handled securely.
- Privacy of users is fully maintained.

8. Maintainability

- The system is built in a modular way.
- Each part (scraping, ML, preprocessing) can be updated separately.
- Easy to improve or replace ML models.
- System upgrades do not affect overall working.

9. Availability

- The system is available anytime through a web browser.
- Users can access it whenever needed.
- Downtime is minimal.
- System recovers quickly after updates or failures.

10. Response Time

- The system responds quickly to user requests.
- Sentiment results are generated in a few seconds.

- Processing is optimized for fast output.
- No long waiting time for users.

11. Fault Tolerance

- The system handles errors like invalid URLs or missing data.
- It does not crash during unexpected situations.
- Proper error messages are shown to users.
- System continues working normally after minor failures.

4.3 System Constraints:

Constraints on the system indicate the limitations and conditions that govern the functioning of the system. Such constraints can impact the effectiveness and accuracy of the sentiment analysis system.

1. Web Scraping Limitations

The system relies on external websites of e-commerce stores in order to gather customer reviews. When there is any change in the structure of the webpage or when a particular website prevents the process of web scraping, the system will not be able to extract information.

2. Data Quality Issues

Sentiment analysis is highly dependent on the quality of the review dataset. In case the data includes spam reviews, extremely short sentences, or vague phrases, it is unlikely that the output will be accurate. Poor or irrelevant data may impact the entire process negatively.

3. Model Limitations

Models like Logistic Regression and Naive Bayes operate according to specific patterns observed in training data. Such models can have difficulty recognizing sarcastic statements, jokes, or any other complicated emotions.

4. Internet Dependency

This system needs a reliable internet connection in order to perform web scraping and access the online data. In case there is any difficulty with the speed of the internet, this will not allow the system to fetch the reviews or return any result.

5. Limited Context Understanding

This model performs text analysis with respect to certain word patterns but lacks the capability of understanding any deep human emotion or context. In this way, the real meaning of certain complicated sentences might not be understood by this system.

6. Fixed Dataset Dependency

This system relies on certain datasets that have been used to train the machine learning models. Due to such dependency, this system may not work efficiently on all kinds of product reviews.

7. API and Website Restrictions

There are some e-commerce sites where automated scraping or any form of permission is required to retrieve information. This will hinder the system from collecting reviews easily from any site

8. Processing Time for Large Data

When the system is processing a lot of data concurrently, the processing time increases. The process of web scraping and sentiment prediction becomes slow with large amounts of data, especially when there is inadequate computing power.

9. Language Limitation

The system mainly works well when dealing with reviews in the English language. When there is review content in another language apart from English, the system might not perform sentiment analysis effectively.

Chapter 5

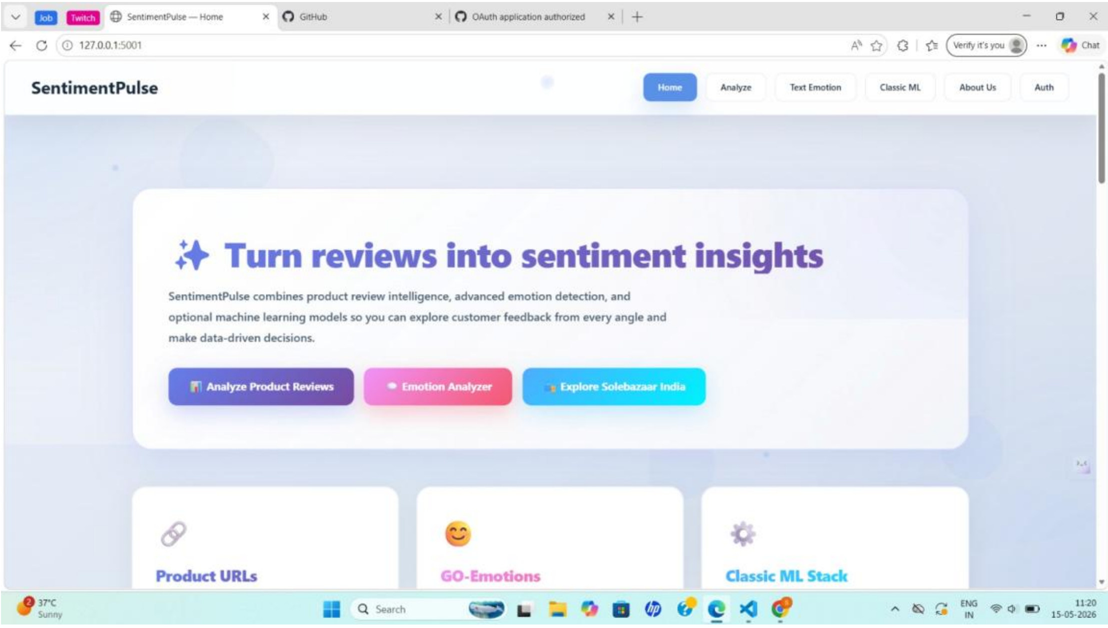
IMPLEMENTATION AND RSEULT

The AI-Powered Customer Sentiment Analysis System for E-commerce uses a systematic process of developing the product, including data processing, natural language processing, machine learning, and web interface components. The AI-Powered Customer Sentiment Analysis system for E-commerce is implemented using the Python language and several supporting libraries such as Pandas, NLTK, and Scikit-learn to facilitate data processing, feature extraction, and training. The process begins by acquiring customer reviews and cleansing them using the preprocessing methods described earlier. Once data has been preprocessed, the next step is to apply the TF-IDF method for converting text into numbers. Then, training is done using a Logistic Regression model. The implementation is done using Flask for the back-end and HTML/CSS for the front-end.

The performance of the proposed system clearly indicates that the developed model is indeed able to accurately categorize customers' feedback into positive, negative, and neutral sentiments. The assessment of performance measures like accuracy, precision, recall, and F1 score illustrates the reliability of the predictions made by the model. Furthermore, the system manages to process customer input in real-time and provide sentiment outputs instantly via the online portal. The prediction results can also be analyzed graphically using basic visualization tools to gain insights into customers' feedback and determine ways to improve business strategies. The results indicate that sentiment analysis can indeed be successfully implemented in e-commerce using machine learning techniques.

5.1 Home page :

The homepage of the AI-powered Customer Sentiment Analysis software (SentimentPulse) features a neat and straightforward layout that allows for easy and convenient navigation around the application. It represents the starting page of the entire system where users learn about its primary task, namely turning customer comments into sentiment analysis data. The navigation bar offers options including Home, Analyze, Text Emotion, Classic ML, About Us, and Authentication, ensuring a smooth transition from one feature to another. The center part of the page displays the main application functions via links to “Analyze Product Reviews” and “Emotion Analyzer” which let users instantly perform sentiment analysis using the software. Moreover, on the homepage, users can see a list of important software features such as analysis of products URL's, emotion detection, and application of advanced machine learning algorithms including TF-IDF and Logistic Regression among others



.Fig: 5.1 Home Page

5.2 About Page:

On the About page of SentimentPulse platform, a comprehensive description of the aims and functions of this application is provided. In particular, it states that SentimentPulse is meant to facilitate a more efficient understanding of customer feedback on products by transforming the product review information into useful sentiment data. The system will help users easily determine which product aspects should be improved and which work perfectly fine. In addition, one can test some of the features of this platform, such as the analysis of product links even without an account, and then register and have all available capabilities at his disposal. The page outlines several major purposes for the use of this tool, including facilitating data-based decisions by users, automatic categorization of reviews into three sentiment types (positive, negative, and neutral), and determining the needs of customers.

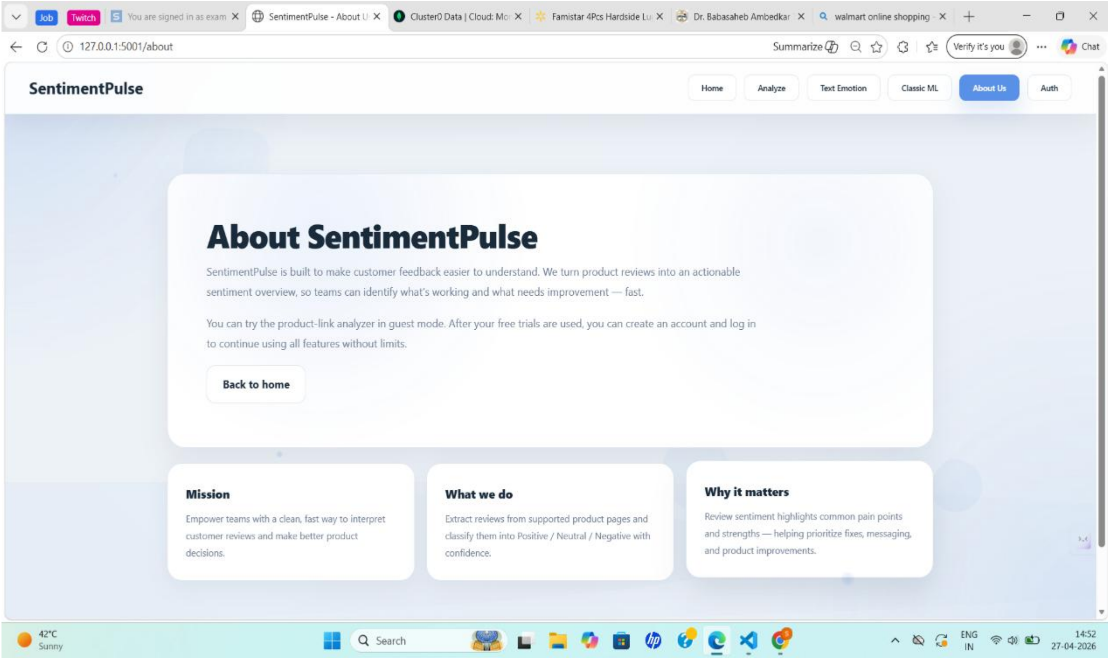


Fig: 5.2 About Page

5.3 : Sign Up Page:

The Sign Up page of the SentimentPulse system enables users who wish to join the application to register for account creation and access. This page gives new users a simple and neat form that helps them fill in their personal details such as names, email address, and password. The page ensures that passwords are confirmed before submission to ensure proper registration. Instructions on how to create passwords and what requirements should be met help new users in generating strong passwords. Users only need to provide their relevant personal information before clicking on the 'Sign

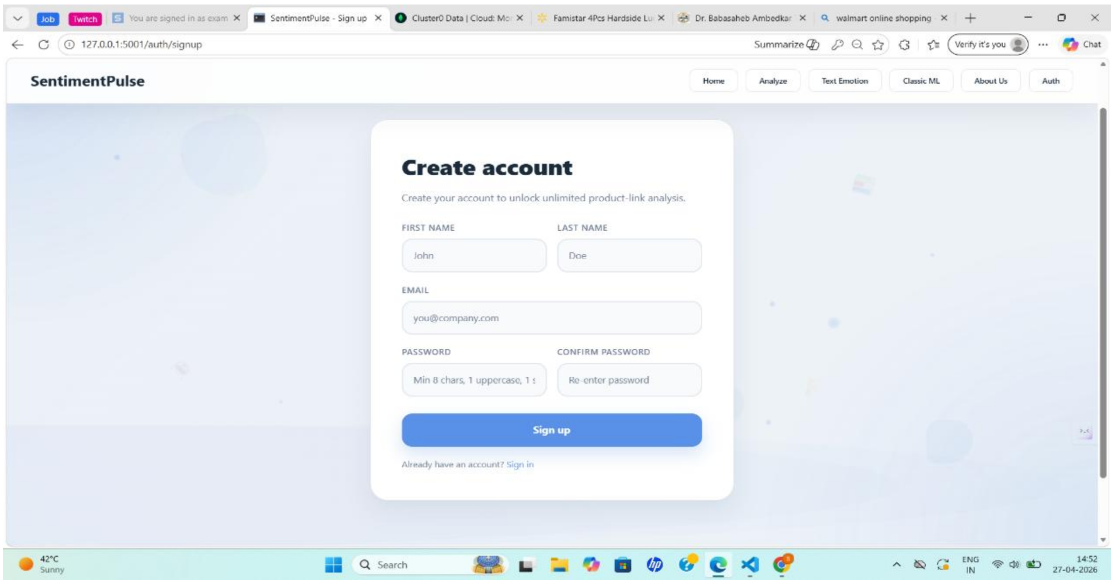


Fig: 5.3: Sign Up Page

5.4 Login Page :

The Login page of the SentimentPulse software is meant for registered users who will be logging into their accounts through their email and password details. It offers a straightforward interface where the user will input his email address and password to gain access into the system. This interface guarantees the security of account access since it will confirm the authenticity of information provided by the user against stored user information. Once logged in, the user can proceed to utilize various functionalities such as sentiment analysis, product reviews, and dashboard functionalities.

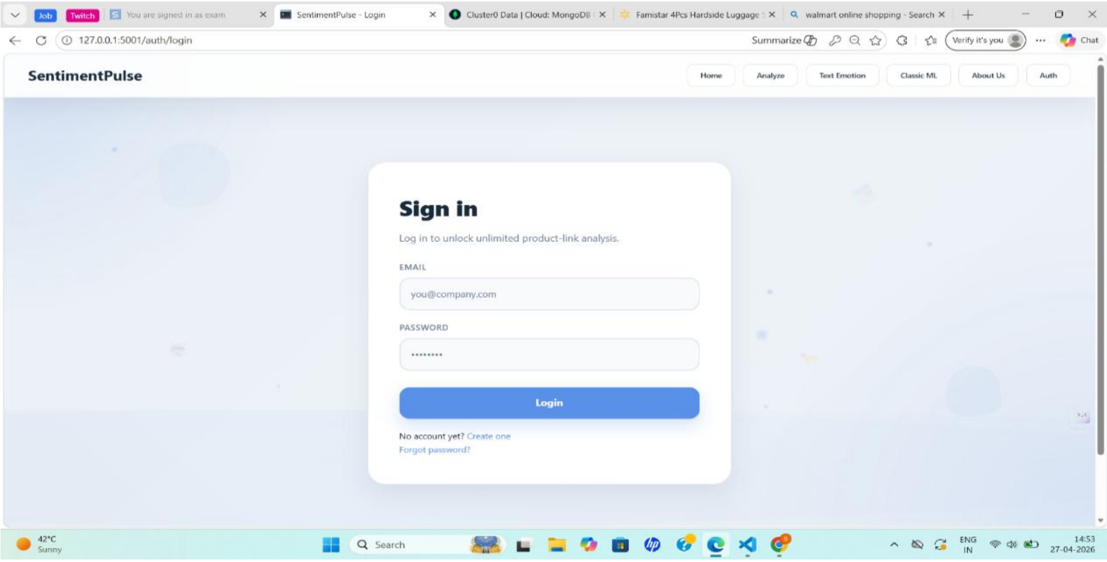


Fig 5.4 : Login Page

5.5 Contact Us Page:

Contact section on the SentimentPulse website ensures that users have an easy way to communicate in case of any doubts or problems. In this regard, the users need to provide their email address and password as their identity details, alongside a message box where they can specify the details of their questions or feedback. This helps the concerned team to comprehend the user's problem. The process is straightforward since users are able to communicate easily through the page. In this regard, the users need to provide their email address and password as their identity details, alongside a message box where they can specify the details of their questions or feedback.

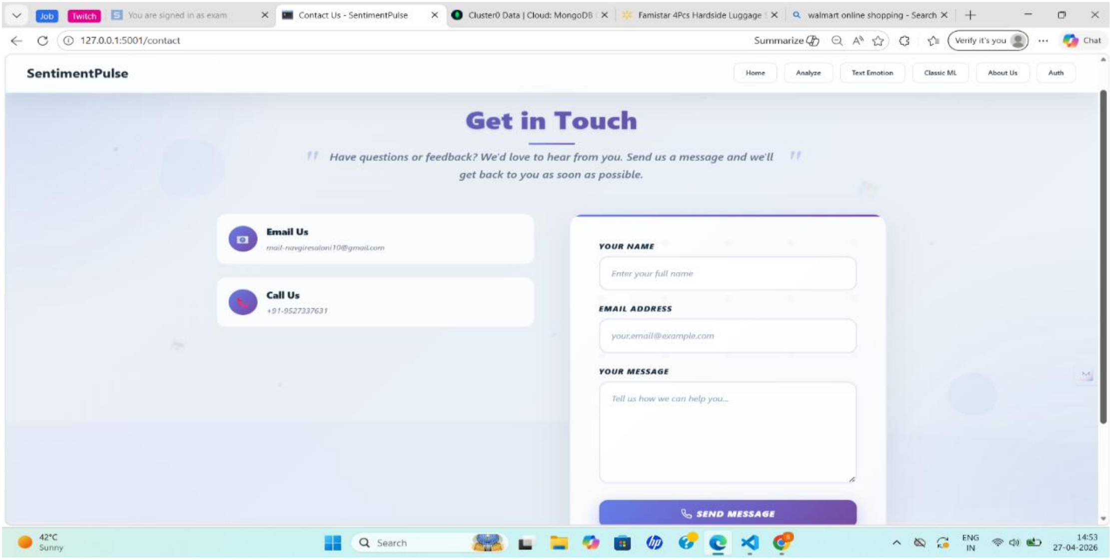


Fig 5.5: Contact Us Page

5.6 Text Emotion Page:

The Text Emotion page on the SentimentPulse platform enables users to have a more intricate analysis of the emotions behind any text beyond basic sentiment analysis. In this case, the user enters any text or review on the provided platform where NLP technologies are used to analyze the emotions behind that text. The resulting page will then show the predominant emotion detected in the text alongside the sentiment category in which the text falls (either positive, negative or neutral). Moreover, the page shows more specific outputs such as happy, sad, disappointment, anger, among others, thus enabling a deeper emotional analysis.

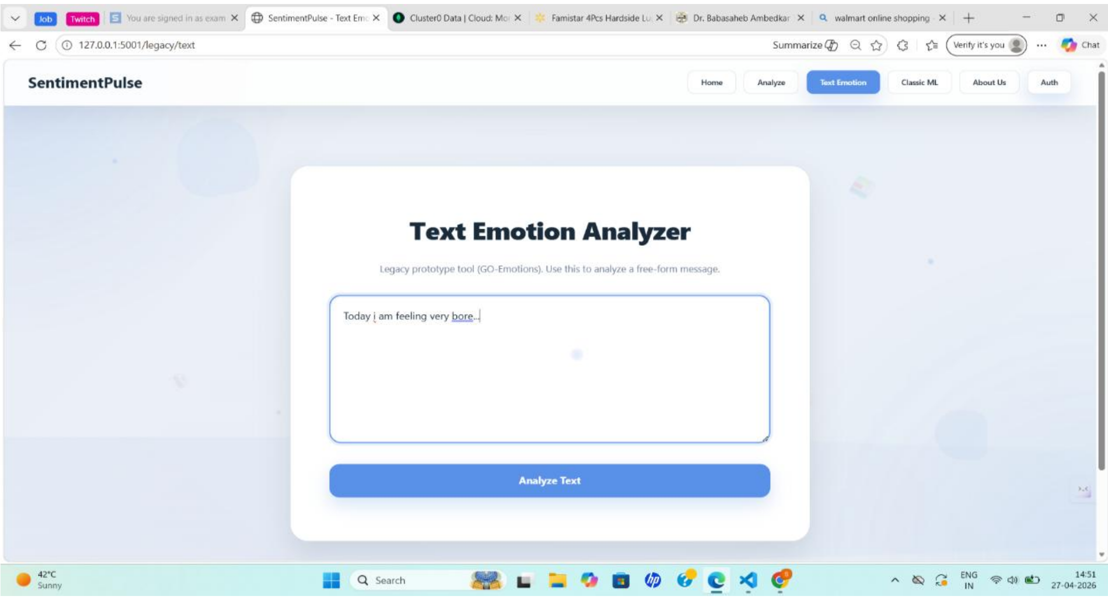


Fig 5.6.1: Text emotion

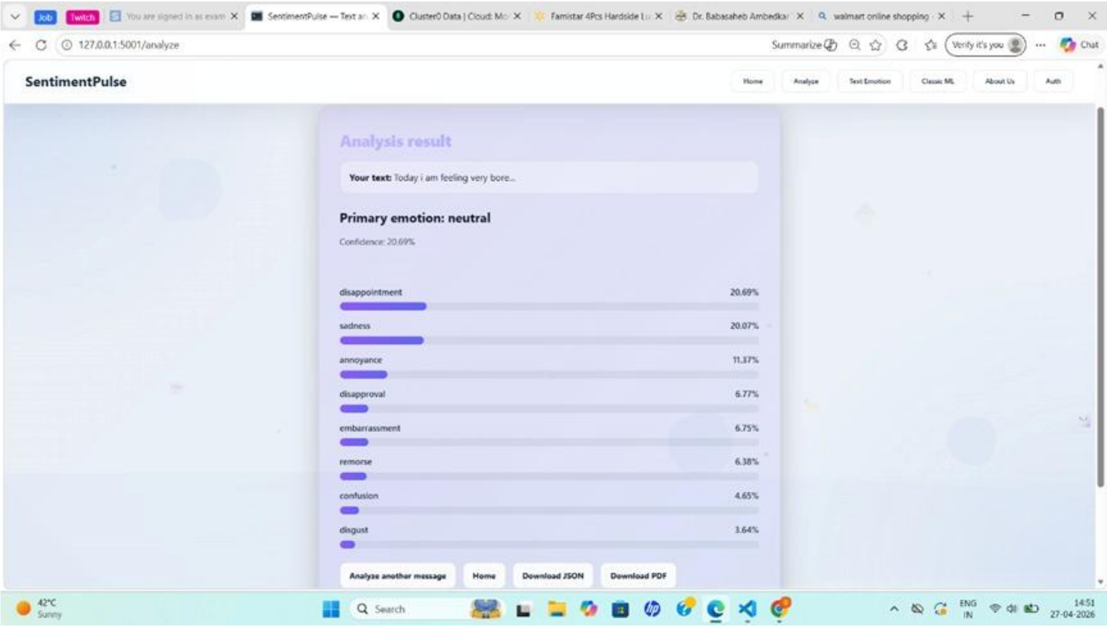


Fig 5.6.2 :Text Emotion Result

The Analytics Dashboard for Emotion Detection provides a comprehensive overview of the system's performance and user activity through clear visual insights. It displays key metrics such as the total number of analyses performed, average confidence score, most frequently detected emotion, and the number of unique emotions identified. The dashboard also includes graphical representations like an emotion distribution chart, which shows the proportion of different emotions detected from the input data, and a confidence trend graph that tracks how prediction confidence changes over time.

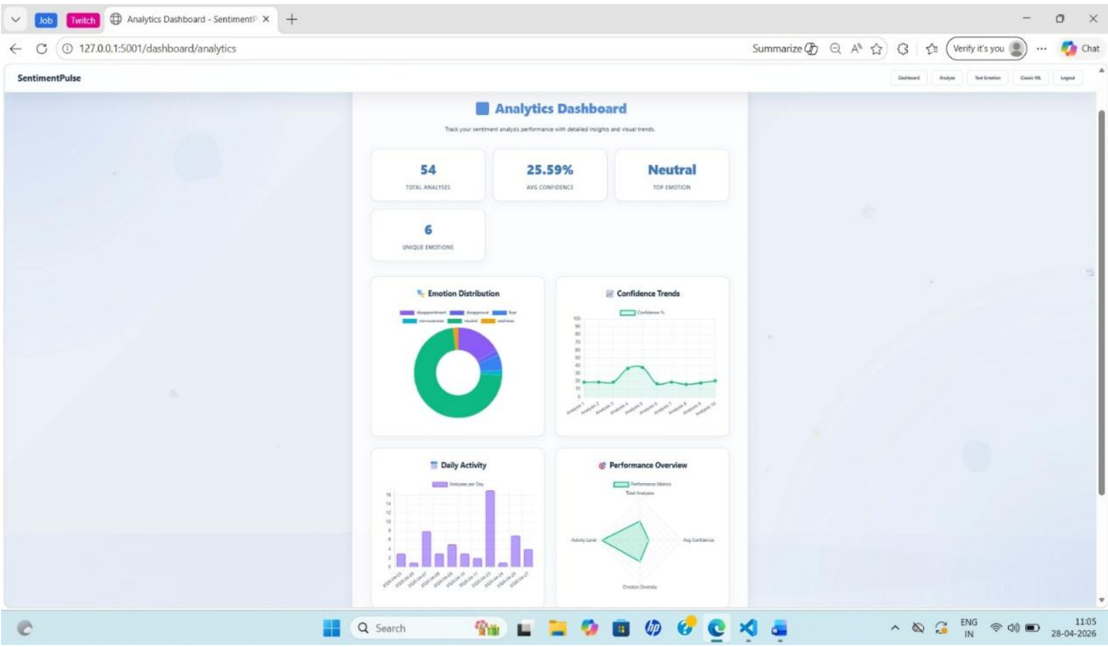


Fig 5.6.3: Text Emotion Dashboard

5.7: Classic ML Page

The Classic ML module of the Sentiment Analysis system offers an intuitive user interface that helps in performing analysis of customer reviews through traditional machine learning algorithms. There are two ways in which users can provide their inputs to this system on this page; one option involves entering the product review in the textbox and another involves providing the product URL from shopping websites like Amazon and Flipkart. Once the input data is entered by the user, this system analyzes the data by deploying the machine learning model that was pre-trained in scikit-learn. The input data to the algorithm is facilitated by means of a vectorizer, which helps convert textual data to numerical representation through TF-IDF method.

When the user enters the input and clicks on the 'Analyze' button, there are several processes that happen in the background. The first process involves cleaning and preprocessing of the entered input data or reviews scraped from the website entered by the user. Unwanted items, such as stop words, punctuations, and special characters, are stripped away. Afterward, the cleaned-up data is converted to numerical vectors and is fed into the Logistic Regression model for classification purposes. Positive, negative, and neutral reviews are then identified based on the predicted results of the classifier.

The Results page gives the user an organized representation of the results of the analysis process. In this section, users can see the sentiment score, average sentiment score, distribution of positive, neutral, and negative reviews in terms of percentage.

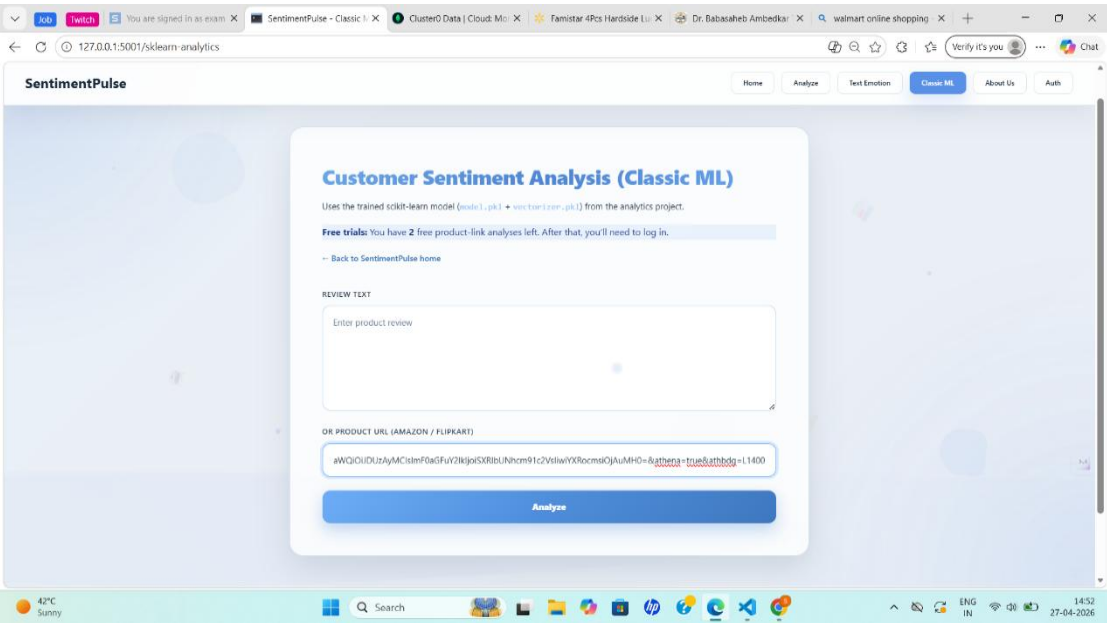


Fig 5.7.1: Classic ML Analysis

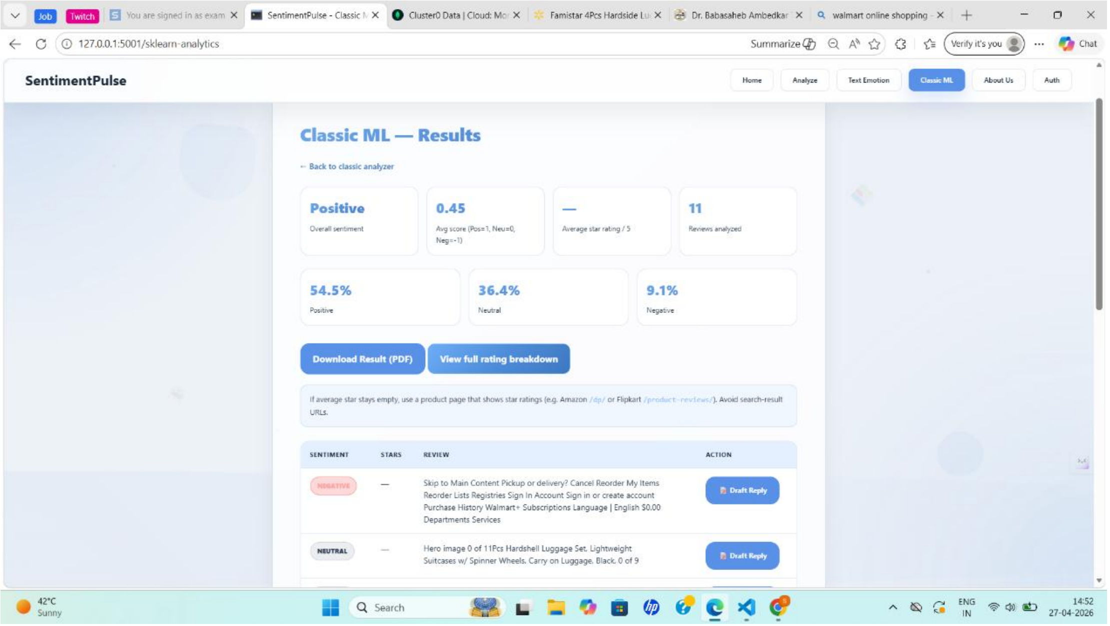


Fig 5.7.2: Classic ML Result

CONCLUSION

The system called "AI-Powered Customer Sentiment Analysis" created in this research work represents a proper way to analyze customers' opinions based on machine learning and NLP. The goal of the research work is focused on classifying the given text data regarding sentiments – positive, negative, and neutral. To convert raw data into machine-readable form, one should employ such steps as text cleaning and feature extraction with the help of TF-IDF. Furthermore, the application of Logistic Regression helps to perform sentiment classification efficiently.

Alongside with back-end processing, one should consider front-end design in order to make the proposed system more convenient for end-users. For instance, the project utilizes such technologies as HTML, CSS, and Flask to implement a proper web-based UI. A user simply inputs the necessary data such as a review or a product URL, and then receives the result immediately after performing required calculations. The output includes not only sentiment but also such factors as sentiment distribution, average scores, and detailed information about the given review. Furthermore, one should note the analytics dashboard with graphics.

The project manages to realize its objective of creating an effective sentiment analysis tool that can be employed by e-commerce sites. The project saves time and labor and increases efficiency in customer opinion assessment. In order to increase the effectiveness of this project, additional improvements can be done including the adoption of more sophisticated models such as deep learning algorithms, multi-language support, and more contextual awareness.

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